

ONDŘEJ THEINER PH.D.

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<https://otheiner.github.io>

SUMMARY

Physicist turned AI evaluation researcher. Five years at CERN executing complex data analyses and developing detector DAQ systems; now applying the same experimental rigor to frontier LLM evaluation — building challenging multimodal tasks, designing contamination-resistant benchmark, and developing novel evaluation methodology.

EXPERIENCE

LLM evaluation specialist (physics) | Mercor / Outlier.ai (remote) 2026 - present

- RLHF - evaluated LLM reasoning on physics tasks for the world leading AI labs
- Created challenging multimodal tasks combining scientific reasoning, programming and data analysis
- Collaborated with a large, cross-disciplinary team of academics

Analysis of the LHC collision data | CERN 2021 - 2024

- Searched for the new physics in the large data sets from the ATLAS experiment
- Used statistical hypothesis testing methods and tools such as ROOT, Python, C++
- Preserved analysis workflow using Docker and RECAST (YAML / framework)

Optimisation of data reconstruction software | CERN 2021 - 2023

- Optimised part of algorithm used for triggering at the ATLAS experiment
- Saved 1.2 % of CPU time used for high level trigger online data reconstruction

Presentation of research results | University of Geneva, CERN 2019 - 2024

- International conferences (ICHEP, ESP-HEP,...) and collaboration meetings

TDAQ development & detector simulation | CERN / Charles University 2016 - 2021

- Developed C++ software for detector readout and DAQ for the FASER experiment
- Simulated and tested response of silicon detectors for CERN experiments

PROJECTS

Benchmarking framework for AI Agents | Personal project 2026 - present

- Procedural benchmarking framework for scientific data analysis tasks
- Introducing metarubrics - novel approach of grounded rubrics by construction
- <https://otheiner.github.io/PARAMETR-Bench-blog/>

SKILLS

Programming: Python (numpy, scikit-learn, matplotlib, scipy, litellm...), C++, R

Tools: shell, git, ROOT, Docker, GitLab CI/CD, L^AT_EX, SQL

Operating systems: Linux, macOS, Windows

Languages: Czech (native), English (C1/C2), French (A1/A2), Spanish (A2)

EDUCATION

Doctoral degree (Ph.D.) - Particle Physics Geneva, Switzerland

University of Geneva, Faculty of Science 2019 - 2024

- In collaboration with CERN, experiments FASER and ATLAS
- Research area: Searches for Physics Beyond Standard Model

Master's degree (Mgr.) - Nuclear and Subnuclear Physics Prague, Czechia

Charles University, Faculty of Mathematics and Physics 2017 - 2019

Bachelor's degree (Bc.) - General Physics Prague, Czechia

Charles University, Faculty of Mathematics and Physics 2014 - 2017

OTHER

ACTIVITIES

- Committee member of the Astronomy Olympiad in the Czech Republic (educational scientific competition for high school students)
- In 2025 walked 3000 km across New Zealand in 117 days on Te Araroa trail